

Design and Analysis of Interaction Method to Adjust Magnification Function Using Microgestures in VR or AR Applications

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ABSTRACT

Among various interaction modes in VR/AR, microgestures have distinct advantages over controllers by reducing fatigue and improving the efficiency. In this paper, we proposed a microgesture set that utilizes the number of fingers as input instructions for magnification adjustment in VR environments. To evaluate the effectiveness of the proposed microgestures, a series of tasks have been designed in three scenes. The results revealed a significant improvement in completion time and subjective measures compared to the performance of controller. These findings offer valuable insights for future gesture design, contributing to the development of more efficient and user-friendly interaction techniques in VR.

Keywords: microgesture, Virtual Reality, ergonomics.

Index Terms: Human-centered computing—Human computer interaction (HCI)—HCI design and evaluation methods—User studies

1 INTRODUCTION

Gesture interaction is extensively employed in VR/AR. However, the design of sensible gestures is crucial. When performing gestures involving arm movements, it can cause hand-arm and shoulder fatigue, posing challenges for prolonged usage [1]. Therefore, microgestures with a smaller range of motion offers distinct advantage over controllers and mid-air gestures.

VR/AR devices provide designers with the means to materialize their ideas [2], but the repetitive creation of identical components increases the interaction task, negatively affecting the interaction experience. A potential solution to enhance interaction efficiency is to use finger count as a representation of the number of copied virtual objects. This approach could reduce users' cognitive load and foster a more natural interaction. Furthermore, the conversion of finger counts to parameters can be applied in various scenarios.

In this paper, we introduce a microgesture set designed for mixed reality tasks by considering user interaction habits and the principles of human factors. The microgesture set utilizes the number of fingers as input instructions. Four types of microgestures were selected for three distinct tasks: a roaming task, a double-speed playback task, and a repeated object creating task. The performance evaluation of the four types of microgestures and the controller encompassed ease of use, user preference, fatigue and completion time. The insights gained from this analysis of microgestures will provide valuable guidance for the design of

microgestures that effectively represent input instructions through finger counts.

2 METHOD

2.1 Apparatus

The virtual scenes were developed by Unity. Leap Motion was used to track users' hand gestures. A desktop (Intel Core i7@3.8GHz, NVIDIA RTX3060, 32GB of RAM) was employed to render the virtual scenes in an HTC Vive Pro.

2.2 Microgestures

The design of these four categories of gestures is grounded on the previous research by Li et al [3]. These gestures have been carefully selected to facilitate users in effortlessly performing multi-finger movements. The microgestures were designed by minimizing fatigue and optimizing usability, adhering to ergonomic principle. As illustrated in Figure 1, four types of microgestures include: (1) Rightward scroll of the finger with the pronated forearm; (2) Scratching motion of the finger toward the pronated forearm; (3) Tapping motion of the finger with the pronated forearm; (4) Forward pointing motion of the finger with the pronated forearm.

A pilot study was conducted to select the preferred finger combination when different numbers of fingers were extended. The results indicated a high degree of consistency in the choice of fingers. Based on these findings, extending the index finger signifies 1×, using the index finger and middle finger represents 2×, employing the middle finger, ring finger and little finger represents 3×, using four fingers from the index finger to little finger represents 4×, and using all five fingers represents 5×.

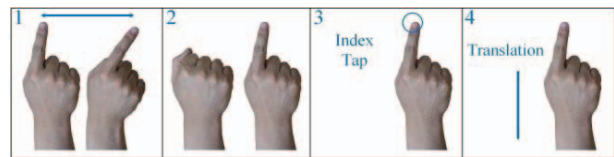


Figure 1: The graphical description of four microgestures.

2.3 Tasks

We designed three scenarios to evaluate the performance of the designed microgestures including roaming, double-speed playback, and repeated object creating. Users can use microgestures and controllers to adjust the forward speed of the car, the playback speed of the music, and the number of objects created in these three types of scenes, respectively. The adjustment rate of each task can be controlled by the controller and the number of fingers involved in microgestures.

2.4 Procedure

Firstly, participants completed tasks in three scenes using four types of microgestures and the controller. Each scene successively presented eight randomly generated commands, and participants were required to complete the corresponding tasks to adjust

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magnification rate according to the prompts. The entire experiment took approximately 15 minutes. Subsequently, they were required to fill in a questionnaire consisting of three subjective measures. The first two measures (Excellent Control and No Hand-Arm Fatigue) were rated on a scale of 1-5, with 1 corresponding to 'strongly disagree' and 5 corresponding to 'strongly agree'. Regarding the preference evaluation, participants ranked the four

microgestures and the controller based on their personal preference, assigning 1 to their favorite and 5 to their least favorite option.

A total of 16 participants (10 males and 6 females), with the age between 22 to 26 years (Mean =23.06, SD =1.06), participated in the study. Among these participants, two (12.5%) had no previous experience with VR, eight (50%) had some experience with VR, and six (37.5%) considered themselves familiar with VR.

Table 1: The summary of subjective and objective measures using four microgestures and the controller for roaming task.

	Gesture 1	Gesture 2	Gesture 3	Gesture 4	Controller	p-Value
Excellent Control	3.9 (0.4)	3.8 (0.4) ^a	4.5 (0.4) ^{ab}	4.1 (0.5)	3.8 (0.4) ^b	0.0170
No Hand-Arm Fatigue	4.3 (0.3) ^a	4.4 (0.4) ^b	4.8 (0.2) ^c	3.8 (0.6) ^{bcd}	2.7 (0.5) ^{abcd}	<0.0001
Preference	4.2 (0.6) ^a	2.8 (1.2) ^{ab}	1.9 (0.6) ^{abc}	1.6 (0.5) ^{abd}	4.5 (0.4) ^{bcd}	<0.0001
Time(s)	34.4 (5.2) ^a	33.8 (5.4) ^b	33.9 (3.9) ^c	34.6 (5.7) ^d	50.5 (7.5) ^{abcd}	<0.0001

Table 2: The summary of subjective and objective measures using four microgestures and the controller for double-speed playback task.

	Gesture 1	Gesture 2	Gesture 3	Gesture 4	Controller	p-Value
Excellent Control	4.2 (0.4)	3.8 (0.2) ^a	4.6 (0.4) ^{ab}	3.9 (0.4) ^b	4.3 (0.3)	0.0017
No Hand-Arm Fatigue	4.0 (0.5) ^a	4.4 (0.3) ^b	4.9 (0.1) ^{ac}	3.9 (0.3) ^{cd}	2.5 (0.5) ^{abcd}	<0.0001
Preference	3.1 (1.3) ^a	2.6 (0.9) ^b	1.5 (0.7) ^{abc}	3.8 (1.5) ^{bc}	3.8 (2.0) ^{bc}	<0.0001
Time(s)	32.4 (3.5) ^a	32.2 (3.9) ^b	31.7 (5.1) ^c	31.8(4.0) ^d	42.9 (4.9) ^{abcd}	<0.0001

Table 3: The summary of subjective and objective measures using four microgestures and the controller for repeated object creating task.

	Gesture 1	Gesture 2	Gesture 3	Gesture 4	Controller	p-Value
Excellent Control	3.6 (0.4) ^a	3.8 (0.6) ^b	4.6 (0.3) ^{ac}	3.8 (0.3) ^{cd}	2.9 (0.5) ^{abcd}	<0.0001
No Hand-Arm Fatigue	3.8 (0.3) ^a	4.4 (0.3) ^{ab}	4.6 (0.4) ^{ac}	3.7 (0.4) ^{bcd}	2.5 (0.5) ^{abcd}	<0.0001
Preference	3.3 (1.3) ^a	2.5 (0.8) ^b	1.8 (1.0) ^{ac}	2.6 (2.0) ^d	4.8 (0.3) ^{abcd}	<0.0001
Time(s)	30.5 (4.5) ^a	30.4 (5.2) ^b	30.2 (6.5) ^c	30.5 (5.6) ^d	41.8 (9.1) ^{abcd}	<0.0001

3 RESULTS

Differences in subjective and objective measures among interaction methods were evaluated using repeated measures ANOVA, and if significant, follow-up pair-wise comparisons were performed using the Tukey test adjusted for multiple comparisons.

The results of four measures for four gestures and the controller were summarized in Tables 1-3. The data was expressed as M(SD), where M is the mean and SD is the standard deviation. Values in a row with the same superscript are significantly different in the follow-up test. Firstly, it could be found that the impact of the interaction method on each measure is found to be significant across all tasks ($p < 0.05$). Specifically, pairwise comparisons revealed that either microgesture significantly outperforms the controller in terms of hand-arm fatigue, mainly attributed to their smaller range of motion. Concerning the subjective measure of excellent control, there is no apparent difference between the controller and microgestures, except for the repeated object creating task. This may be because the repetitive use of the controller to create objects negatively affects users' interactive experience. Notably, the microgestures are overwhelmingly preferred over the controller for performing the all three tasks. Furthermore, the completion time for each task using the controller is significantly longer compared to either microgesture. Consequently, users exhibit a stronger inclination to employ microgestures for interaction rather than controllers.

Gestures with smaller movements help reduce fatigue. Gesture 2 and 3 consistently exhibit less fatigue compared to other microgestures across all three tasks, owing to their minimal range of motion and involvement of only finger movements. Furthermore, among the four microgestures, gesture 3 consistently demonstrates superior control, irrespective of the task. In terms of completion time, there is no significant difference among the four types of microgestures. Additionally, except for the roaming task, gesture 3 was consistently the most preferred choice for completing the

interaction. However, for the roaming task, gesture 4 emerged as the preferred choice, likely due to its alignment with the forward motion of the car in the task.

4 IMPLICATIONS FOR DESIGN AND FUTURE WORK

Based on the results of this study, a microgesture set utilizing the number of fingers as input instructions was proposed for VR /AR system. This approach has the potential to improve the interaction efficiency while minimizing fatigue. The preference for gesture 3 indicates that users are inclined to choose gestures that resemble familiar interaction patterns, such as clicking and sliding when operating a touch device. Moreover, the user preference is also related to the consistency between gestures and specific tasks. For example, in the roaming task, users favored gesture 4 for completing the task, as it aligned well with the task of the car moving forward. While some participants noted that gesture 1 (sliding action) closely resembled steering or adjusting playback progress. In future gesture design, designers should prioritize the consistency between gestures and specific tasks.

In the future, we will explore the performance of more interaction modes to identify the optimal mode for different tasks. This will contribute to a more comprehensive understanding of the diverse options available for improving user interaction.

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